# PROJECT 1 – FSM DESIGNS: MEALY & MOORE MACHINES AND UP-DOWN COUNTER

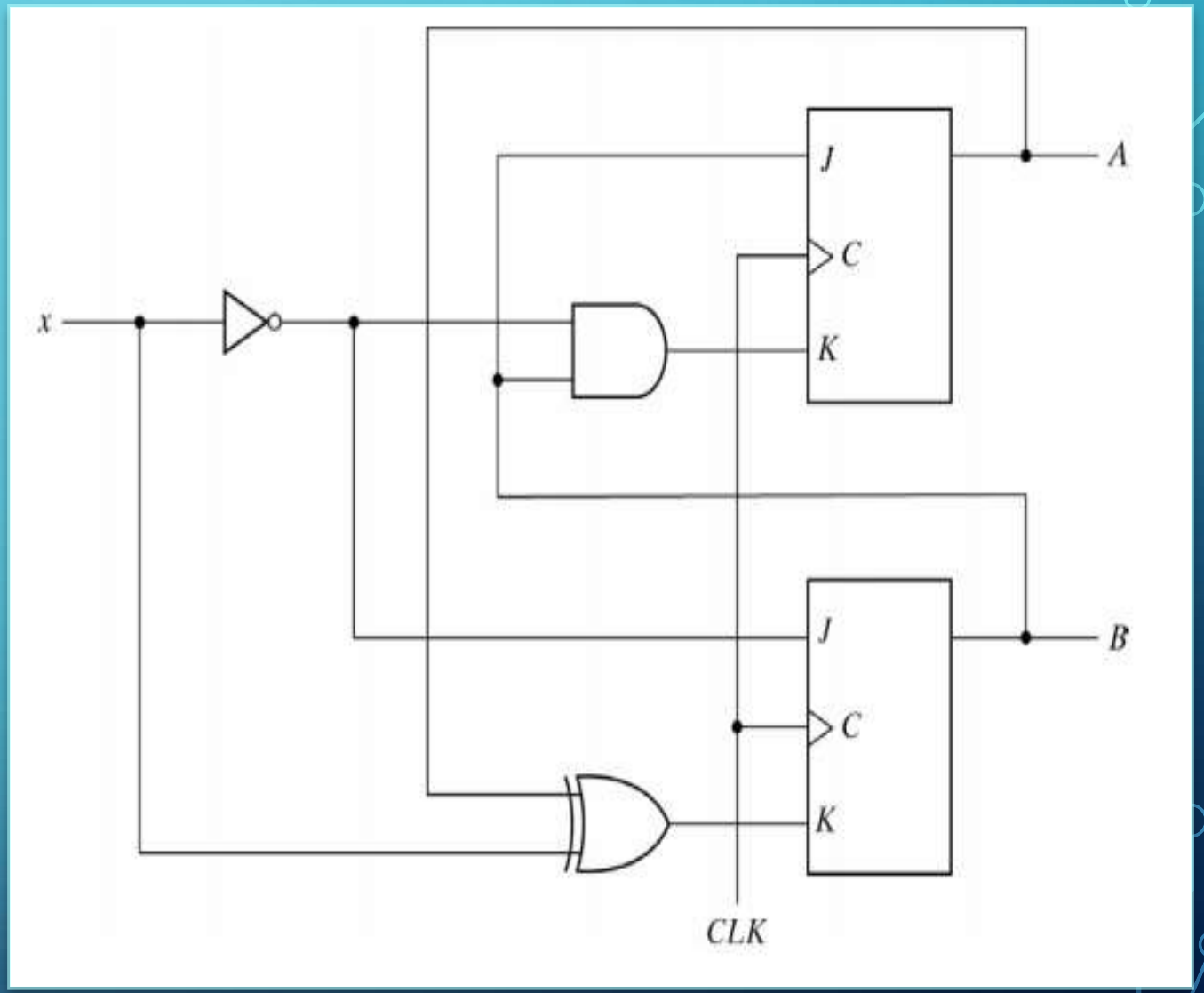
ADAMYA SINGH | **VERY LARGE-SCALE INTEGRATION (VLSI) DOMAIN** |

30/01/2026

# MOORE MACHINES

In a Moore model, the outputs of the sequential circuit are synchronized with the clock because they depend on only flip-flop outputs that are synchronized with the clock

Moore model – The output is a function of the present state only.



C:\Users\Acer\Desktop\VLSI Projects\Moore_machine.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

D_flip_flop_test.v | Up_Down_counter.v | Up_Down_counter_test.v | Melay_machine_test.v | Melay_machine.v | Moore_machine.v | Moore_machine_test.v | JK_flip_flop.v | JK_flip_flop_test.v

```
1  module Moore_machine(
2  input x, clk,
3  output A,B
4  );
5
6  wire xbar, a, b;
7  wire J_A, K_A, J_B, K_B;
8
9
10 not n1(xbar,x);
11 and a1(a,xbar,J_A);
12 xor x1(b,x,A);
13
14 assign J_A = B;
15 assign K_A = a;
16 assign J_B = xbar;
17 assign K_B = b;
18
19
20 JK_flip_flop FF_A (
21         .clk(clk),
22         .J(J_A),    // Connect wire J_A to pin J
23         .K(K_A),    // Connect wire K_A to pin K
24         .Q(A)       // Output goes to wire A
25     );
26
27 JK_flip_flop FF_B (
28         .clk(clk),
29         .J(J_B),    // Connect wire J_B to pin J
30         .K(K_B),    // Connect wire K_B to pin K
31         .Q(B)      // Output goes to wire B
32     );
33
34 endmodule
```

C:\Users\Acer\Desktop\VLSI Projects\Moore_machine_test.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

D_flip_flop_test.v | Up_Down_counter.v | Up_Down_counter_test.v | Melay_machine_test.v | Melay_machine.v | Moore_machine.v | Moore_machine_test.v | JK_flip_flop.v | JK_flip_flop_test.v

```
1   module Moore_machine_test;
2
3   reg x, clk;
4   wire xbar, a, b;
5   wire J_A, K_A, J_B, K_B;
6
7   initial
8       begin
9           clk = 0;
10          forever #3 clk = ~clk;
11      end
12
13  initial
14  begin
15  $dumpfile("Moore_machine.vcd");
16  $dumpvars(0,Moore_machine_test);
17
18  x=0;
19  #10 x=0;
20  #10 x=1;
21  #10;
22  #10 $finish;
23  end
24
25  Moore_machine mac(x,clk,A,B);
26
27
28  endmodule
```

C:\Users\Acer\Desktop\VLSI Projects\JK_flip_flop.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

D_flip_flop_test.v | Up_Down_counter.v | Up_Down_counter_test.v | Melay_machine_test.v | Melay_machine.v | Moore_machine.v | Moore_machine_test.v | JK_flip_flop.v | JK_flip_flop_test.v

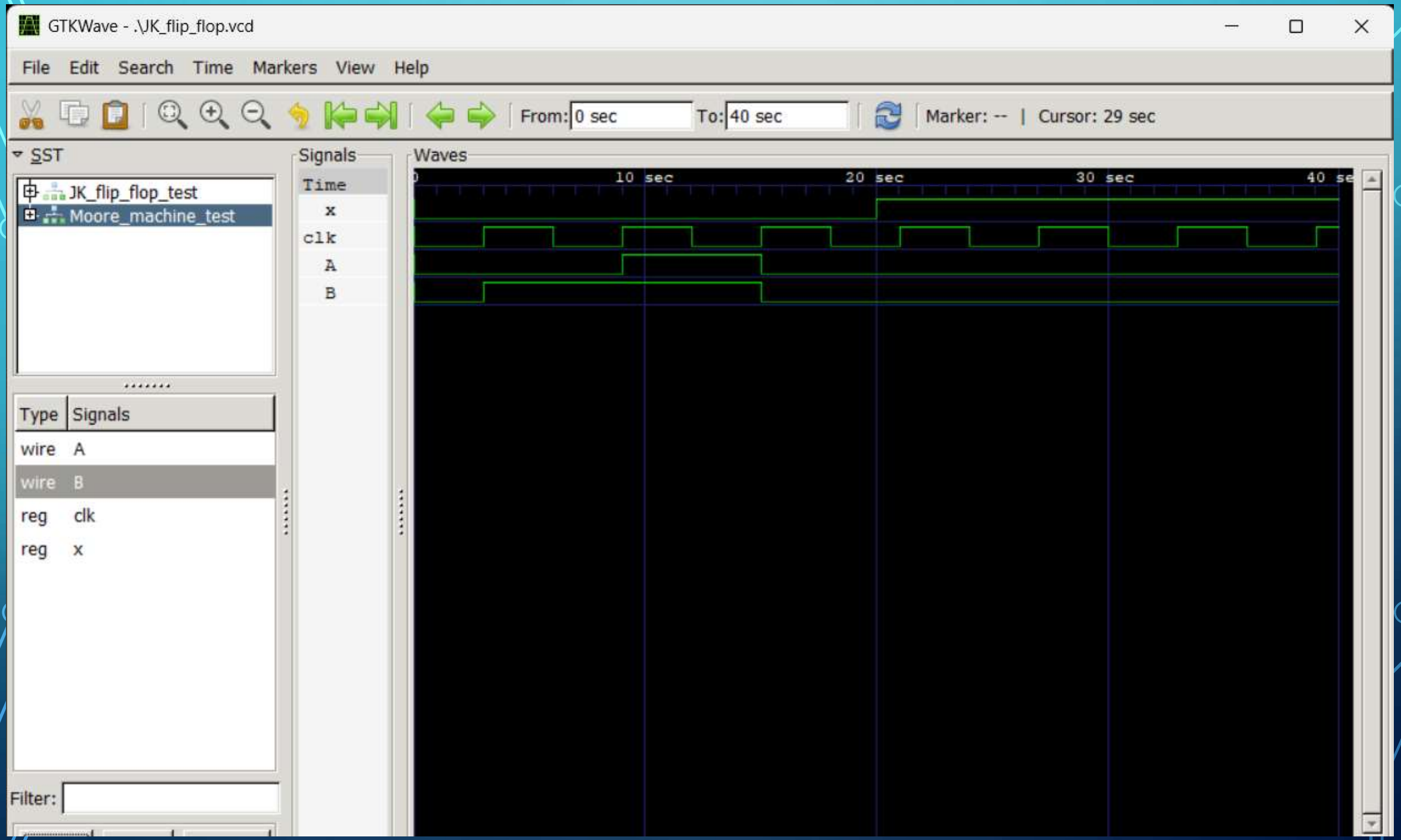
```
1  module JK_flip_flop(
2  input J, clk, K,
3  output reg Q
4  );
5
6  initial Q = 0; // Initialize to 0
7
8      always @(posedge clk) begin
9          case ({J,K})
10             2'b00: Q <= Q;    // Hold
11             2'b01: Q <= 0;    // Reset
12             2'b10: Q <= 1;    // Set
13             2'b11: Q <= ~Q;   // Toggle
14         endcase
15     end
16
17 endmodule
18
```

C:\Users\Acer\Desktop\VLSI Projects\JK_flip_flop_test.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

D_flip_flop_test.v | Up_Down_counter.v | Up_Down_counter_test.v | Melay_machine_test.v | Melay_machine.v | Moore_machine.v | Moore_machine_test.v | JK_flip_flop.v | JK_flip_flop_test.v

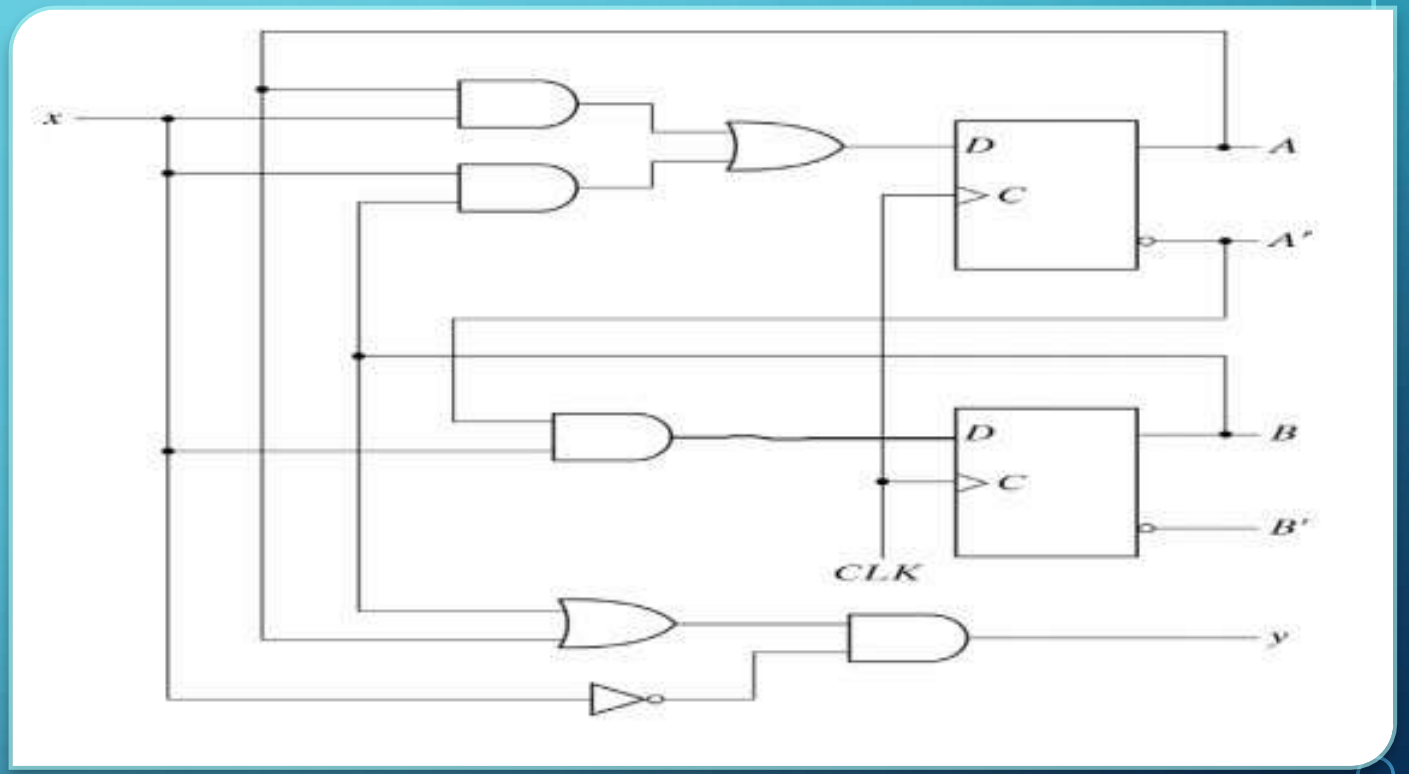
```
1  module JK_flip_flop_test;
2
3  reg J,K,clk;
4  wire a, b, Qbar;
5
6  //CLOCK GENERATION: Toggle cloch every 5 time units
7      initial
8      begin
9          clk = 0;
10         forever #5 clk = ~clk;
11     end
12
13 initial
14 begin
15 $dumpfile("JK_flip_flop.vcd");
16 $dumpvars(0,JK_flip_flop_test);
17
18 J=0; K=0;
19 #10;
20 #10 J=0; K=0;
21 #10 J=0; K=1;
22 #10 J=1; K=0;
23 #10 J=1; K=1;
24 #10;
25 #10;
26 #10;
27 #10 $finish;
28 end
29
30 JK_flip_flop gates(J,K,clk,Q);
31
32 endmodule
```


GTKWave - .\JK_flip_flop.vcd
File Edit Search Time Markers View Help
From: 0 sec
To: 40 sec
Marker: -- | Cursor: 29 sec
SST
JK_flip_flop_test
Moore_machine_test
Type Signals
wire A
wire B
reg clk
reg x
Filter:
Signals
Time
x
clk
A
B
Waves
10 sec
20 sec
30 sec
40 se

# MEALY MACHINES

In a Mealy model, the outputs may change if the inputs change during the clock cycle. To achieve synchronization, the inputs must be synchronized with the clock and the outputs must be sampled only during the clock edge.

Mealy model – The output is a function of both the present state and input.



C:\Users\Acer\Desktop\VLSI Projects\Melay_machine.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

Up_Down_counter.v | Up_Down_counter_test.v | Melay_machine_test.v | Melay_machine.v | D_flip_flop.v | D_flip_flop_test.v

```
1  module Melay_machine(
2  input x, clk,
3  output y, A, B, Abar, Bbar
4  );
5
6  wire a, b, c, o1, o2;
7  wire D_A, D_B;
8
9  and a1(a, x, A);
10 and a2(b, x, B);
11 and a3(c, x, Abar);
12 and a4(y, xbar, o2);
13 or or1(o1, a, b);
14 or or2(o2, A, B);
15 not n1(xbar, x);
16
17 assign D_A = o1;
18 assign D_B = c;
19
20 D_flip_flop FF_A (
21         .clk(clk),
22         .D(D_A),    // Connect wire D_A to pin D
23         .Q(A),       // Output goes to wire A
24         .Qbar(Abar)
25     );
26
27 D_flip_flop FF_B (
28         .clk(clk),
29         .D(D_B),    // Connect wire D_B to pin D
30         .Q(B),      // Output goes to wire B
31         .Qbar(Bbar)
32     );
33
34 endmodule
```

C:\Users\Acer\Desktop\VLSI Projects\Melay_machine_test.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

Up_Down_counter.v | Up_Down_counter_test.v | Melay_machine_test.v | Melay_machine.v | D_flip_flop.v | D_flip_flop_test.v

```
1  module Melay_machine_test;
2  reg x, clk;
3
4  wire a, b, c, o1, o2;
5  wire D_A, D_B;
6
7
8  initial
9      begin
10         clk = 0;
11         forever #5 clk = ~clk;
12     end
13
14 initial
15 begin
16 $dumpfile("Melay_machine.vcd");
17 $dumpvars(0,Melay_machine_test);
18
19 x=0;
20 #10 x=0;
21 #10 x=1;
22 #10;
23 #10 $finish;
24 end
25
26
27 Melay_machine mac(x, clk, y, A, B, Abar, Bbar);
28
29
30 endmodule
```

C:\Users\Acer\Desktop\VLSI Projects\D_flip_flop.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

Up_Down_counter.v | Up_Down_counter_test.v | Melay_machine_test.v | Melay_machine.v | D_flip_flop.v | D_flip_flop_test.v

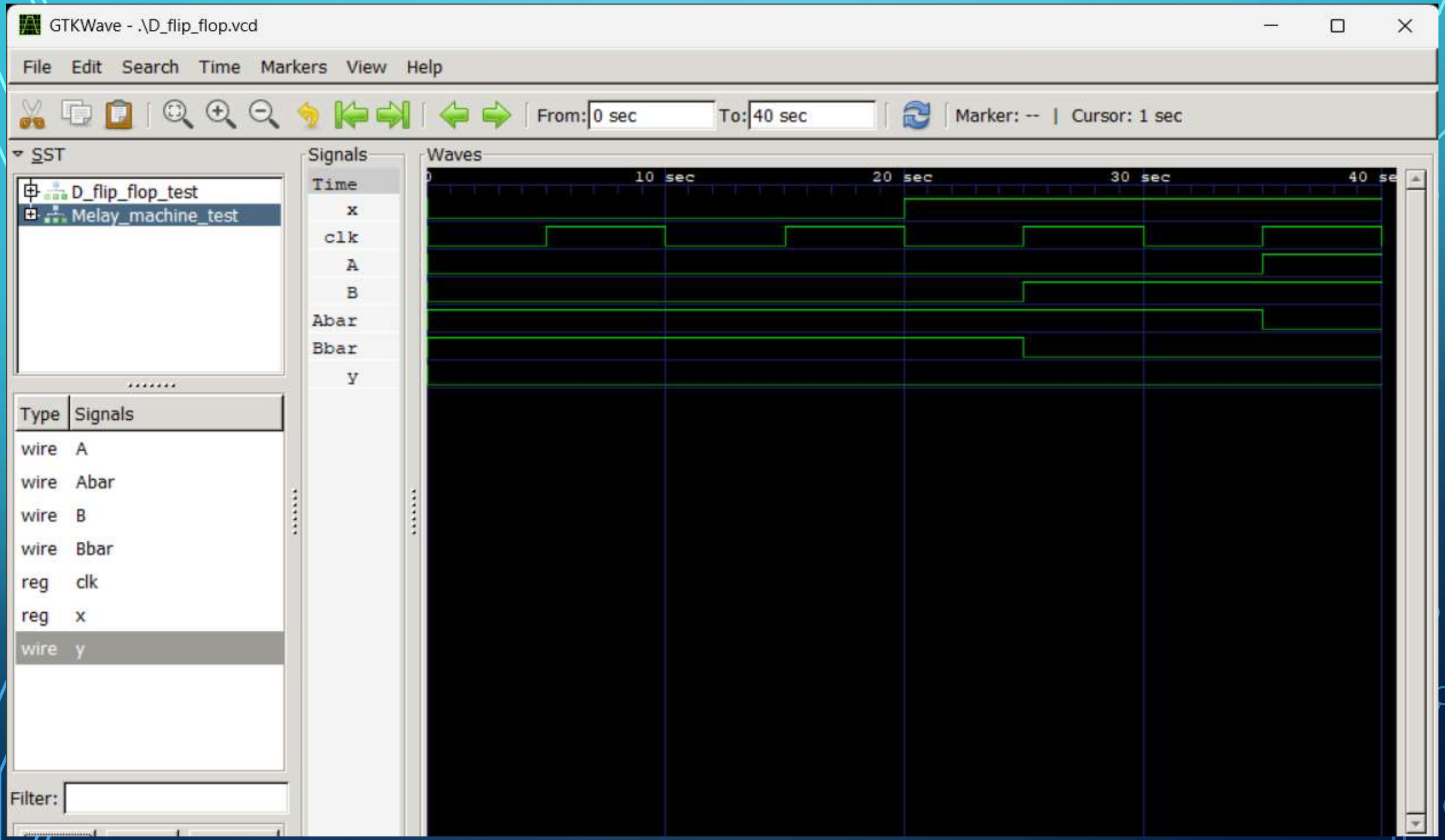
```
1  module D_flip_flop(
2  input D, clk,
3  output reg Q,
4  output Qbar
5  );
6
7  initial Q = 0;
8  assign Qbar = ~Q;
9
10     // The "Camera Snapshot" Logic
11     always @(posedge clk) begin
12         Q <= D;
13     end
14
15 endmodule
```

C:\Users\Acer\Desktop\VLSI Projects\D_flip_flop_test.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

Up_Down_counter.v | Up_Down_counter_test.v | Melay_machine_test.v | Melay_machine.v | D_flip_flop.v | D_flip_flop_test.v

```
1  module D_flip_flop_test;
2  reg D, clk;
3  wire Q, Qbar;
4
5
6  initial
7      begin
8          clk = 0;
9          forever #5 clk = ~clk;
10     end
11
12 initial
13 begin
14 $dumpfile("D_flip_flop.vcd");
15 $dumpvars(0,D_flip_flop_test);
16
17 D=0;
18 #10;
19 #10 D=0;
20 #10 D=1;
21 #10;
22 #10 D=1;
23 #10 D=0;
24 #10;
25 #10 $finish;
26 end
27
28 D_flip_flop gates(D,clk,Q,Qbar);
29
30 endmodule
```

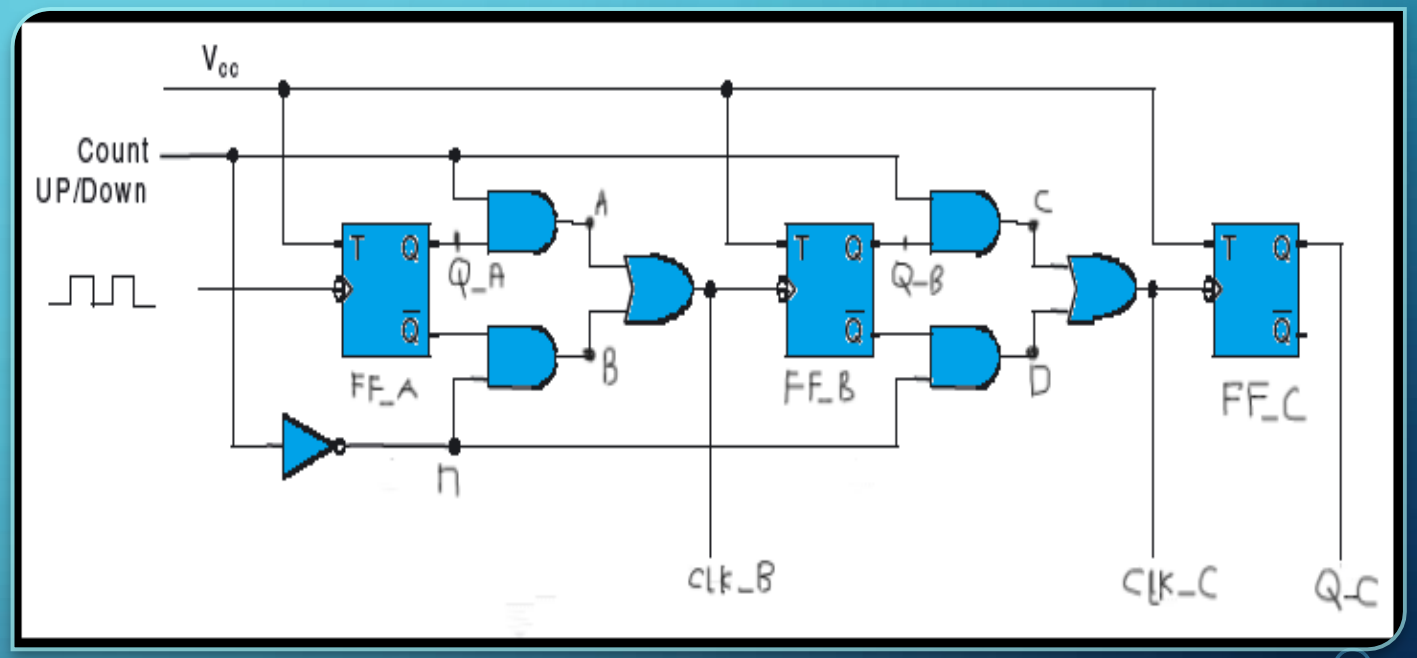

GTKWave - .\D_flip_flop.vcd
File Edit Search Time Markers View Help
From: 0 sec
To: 40 sec
Marker: -- | Cursor: 1 sec
SST
D_flip_flop_test
Melay_machine_test
Type Signals
wire A
wire Abar
wire B
wire Bbar
reg clk
reg x
wire y
Filter:
Signals
Time
x
clk
A
B
Abar
Bbar
y
Waves
10 sec
20 sec
30 sec
40 se

# UP-DOWN COUNTER

This design utilizes an Asynchronous (Ripple) Architecture, where the clock for subsequent flip-flops is derived from the outputs of previous stages rather than a single global clock.

A flip-flop can hold single bit; so, for 3-bit operation the counter shall need 3 flip-flops. An inverter has been inserted in between the count-up/down (mode) control line to ensure that the count-up/down cannot be in the HIGH state simultaneously.

When the count-up/down line is held HIGH, the lower AND gates will be disabled and their outputs will be zero. So, they will not affect the outputs of the OR gates. At the same time the upper AND gates will be enabled. Hence, *clk_B* will pass through the OR gate and into the clock input of the B flip-flop. Similarly, *clk*_c will be gated into the clock input of the C flip-flop. Thus, as the input pulses are applied, it will count up and follow a natural binary counting sequence from 000 to 111.



C:\Users\Acer\OneDrive\Desktop\VLSI Projects\T_flip_flop.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

T_flip_flop.v | T_flip_flop_test.v | Up_Down_counter.v | Up_Down_counter_test.v

```
1  module T_flip_flop(
2  input T, clk,
3  output reg Q,
4  output Qbar
5  );
6
7  initial Q = 0;
8  assign Qbar = ~Q;
9
10 always @(posedge clk)
11     begin
12         Q <= Q^T;
13     end
14
15 endmodule
```

C:\Users\Acer\OneDrive\Desktop\VLSI Projects\T_flip_flop_test.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

T_flip_flop.v | T_flip_flop_test.v | Up_Down_counter.v | Up_Down_counter_test.v

```
1  module T_flip_flop_test;
2  reg T, clk;
3  wire Q, Qbar;
4
5
6  initial
7      begin
8          clk = 0;
9          forever #5 clk = ~clk;
10     end
11
12 initial
13 begin
14 $dumpfile("T_flip_flop.vcd");
15 $dumpvars(0,T_flip_flop_test);
16
17 T=0;
18 #10;
19 #10 T=1;
20 #10 T=0;
21 #10;
22 #10 T=1;
23 #10 T=0;
24 #10;
25 #10 $finish;
26 end
27
28 T_flip_flop gates(T,clk,Q,Qbar);
29
30 endmodule
```

C:\Users\Acer\OneDrive\Desktop\VLSI Projects\Up_Down_counter.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

T_flip_flop.v | T_flip_flop_test.v | Up_Down_counter.v | Up_Down_counter_test.v

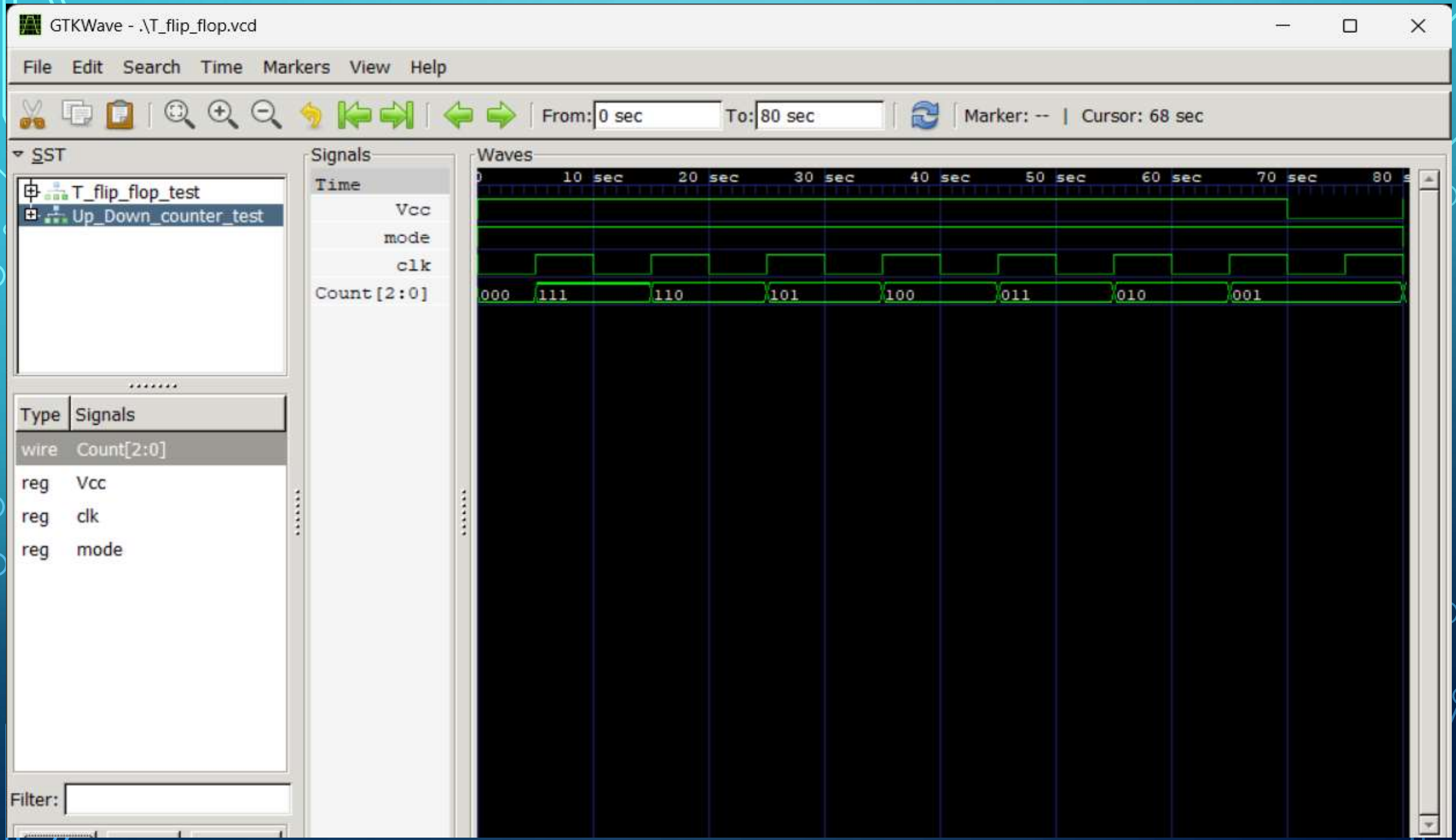
```
1  module  Up_Down_counter(
2  input Vcc, clk, mode,
3  output [2:0] Count
4  );
5
6  wire n, A, B, C, D;
7  wire Q_A, Q_Abar, Q_B, Q_Bbar, Q_C, Q_Cbar;
8  wire clk_B, clk_C;
9  wire T_A, T_B, T_C;
10
11 not n1(n, mode);
12 and a1(A, mode, Q_A);
13 and a2(B, Q_Abar, n);
14 and a3(C, mode, Q_B);
15 and a4(D, Q_Bbar, n);
16 or r1(clk_B, A, B);
17 or r2(clk_C, C, D);
18
19 assign Count = {Q_C, Q_B, Q_A};
20 assign T_A = Vcc;
21 assign T_B = Vcc;
22 assign T_C = Vcc;
23
24 T_flip_flop FF_A (
25         .clk(clk),
26         .T(T_A),    // Connect wire D_A to pin D
27         .Q(Q_A),       // Output goes to wire A
28         .Qbar(Q_Abar)
29     );
30
31 T_flip_flop FF_B (
32         .clk(clk_B),
33         .T(T_B),    // Connect wire D_B to pin D
34         .Q(Q_B),      // Output goes to wire B
35         .Qbar(Q_Bbar)
36     );
37
38 T_flip_flop FF_C (
44 endmodule
```

C:\Users\Acer\OneDrive\Desktop\VLSI Projects\Up_Down_counter_test.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

T_flip_flop.v | T_flip_flop_test.v | Up_Down_counter.v | Up_Down_counter_test.v

```
1  module Up_Down_counter_test;
2
3  reg Vcc, clk, mode;
4  wire [2:0] Count;
5
6  initial
7      begin
8          clk = 0;
9          forever #5 clk = ~clk;
10     end
11
12 initial
13 begin
14 $dumpfile("Up_Down_counter.vcd");
15 $dumpvars(0,Up_Down_counter_test);
16
17 Vcc=1; mode=1;
18 #70;
19
20 Vcc=0;
21 #10;
22
23 Vcc=1; mode=0;
24 #70;
25 $finish;
26 end
27
28
29 Up_Down_counter gates(Vcc, clk, mode, Count);
30
31
32 endmodule
```


GTKWave - .\T_flip_flop.vcd
File Edit Search Time Markers View Help
From: 0 sec
To: 80 sec
Marker: -- | Cursor: 68 sec
SST
T_flip_flop_test
Up_Down_counter_test
Type Signals
wire Count[2:0]
reg Vcc
reg clk
reg mode
Filter:
Signals
Time
Vcc
mode
clk
Count[2:0]
Waves
10 sec 20 sec 30 sec 40 sec 50 sec 60 sec 70 sec 80
000 111 110 101 100 011 010 001

# REFERENCES:


1. Flip-Flop Types – GeeksforGeeks
2. Mealy and Moore Models
3. Asynchronous 3-bit up down counter | Electronics Engineering Study Center
4. Design Asynchronous Up/Down Counter - GeeksforGeeks